

Mini-ERP Platform — User Guide

Product: Mini-ERP Platform
Repository: <https://github.com/deepakumarsray2026-dev/mini-erp>
Live API: <http://34.13.57.203:8000/docs>

Document History

Version	Date	Author	Description
v3.2.0	2026-04-20	deepakkumarsray2026-dev	Phase 3 bug fixes: Finance Chat UX, inline tool-call handling, SQL savepoint isolation
v3.1.0	2026-04-15	deepakkumarsray2026-dev	Phase 3 initial release: Finance Chat, Duplicate Invoice Detection
v2.0.0	2026-03-XX	deepakkumarsray2026-dev	Phase 2: ML models, AI Dashboard (Attrition, Expense Violations, Payroll Anomaly, Invoice Classifier)
v1.0.0	2026-02-XX	deepakkumarsray2026-dev	Phase 1 MVP: Core ERP modules (Workforce, Payroll, AP, Expenses, Procurement, GL)
v0.1.0	2026-01-XX	deepakkumarsray2026-dev	Week 0: GCP infrastructure, Docker Compose stack, CI/CD, DB schema

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1. Platform Overview

Mini-ERP is a full-stack Enterprise Resource Planning platform built as a portfolio project demonstrating a progressive multi-phase AI/ML journey on top of a solid CRUD foundation.

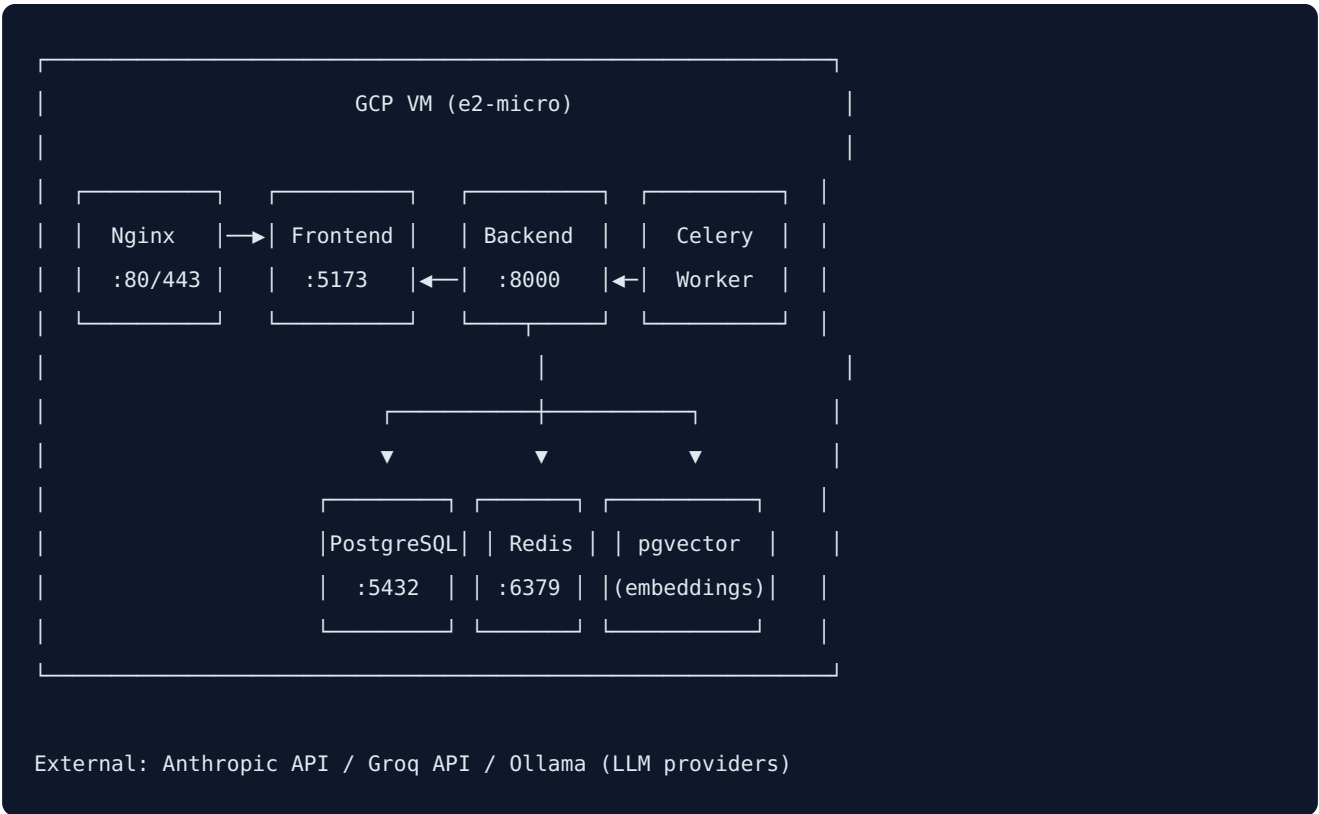
Technology Stack

Layer	Technology
Backend	FastAPI (Python 3.11), SQLAlchemy 2.0, Alembic, Celery
Frontend	React 18, TypeScript, Vite, Tailwind CSS, TanStack Query
Database	PostgreSQL 15 (8 schema namespaces) + pgvector
Cache / Queue	Redis 7
Infrastructure	Docker Compose, GCP <code>e2-micro</code> (free tier), Nginx

Roadmap

Phase	Description	Status
Week 0	Infrastructure (GCP, GitHub, CI/CD, DB schema)	☑ Complete
Phase 1	MVP ERP CRUD (6 modules)	☑ Complete
Phase 2	Machine Learning (4 models)	☑ Complete
Phase 3	RAG + LLM (Finance Chat, Duplicate Invoice Detection)	☑ Complete
Phase 4	Deep Learning (LSTM Budget Forecaster, CNN Invoice Classifier)	☐ Pending
Phase 5	AI Agents (Invoice, Expense Audit, Onboarding)	☐ Pending
Phase 6	Agentic Networks	☐ Pending
Week 11	Polish, Docs, Portfolio Deploy	☐ Pending

2. Architecture Summary



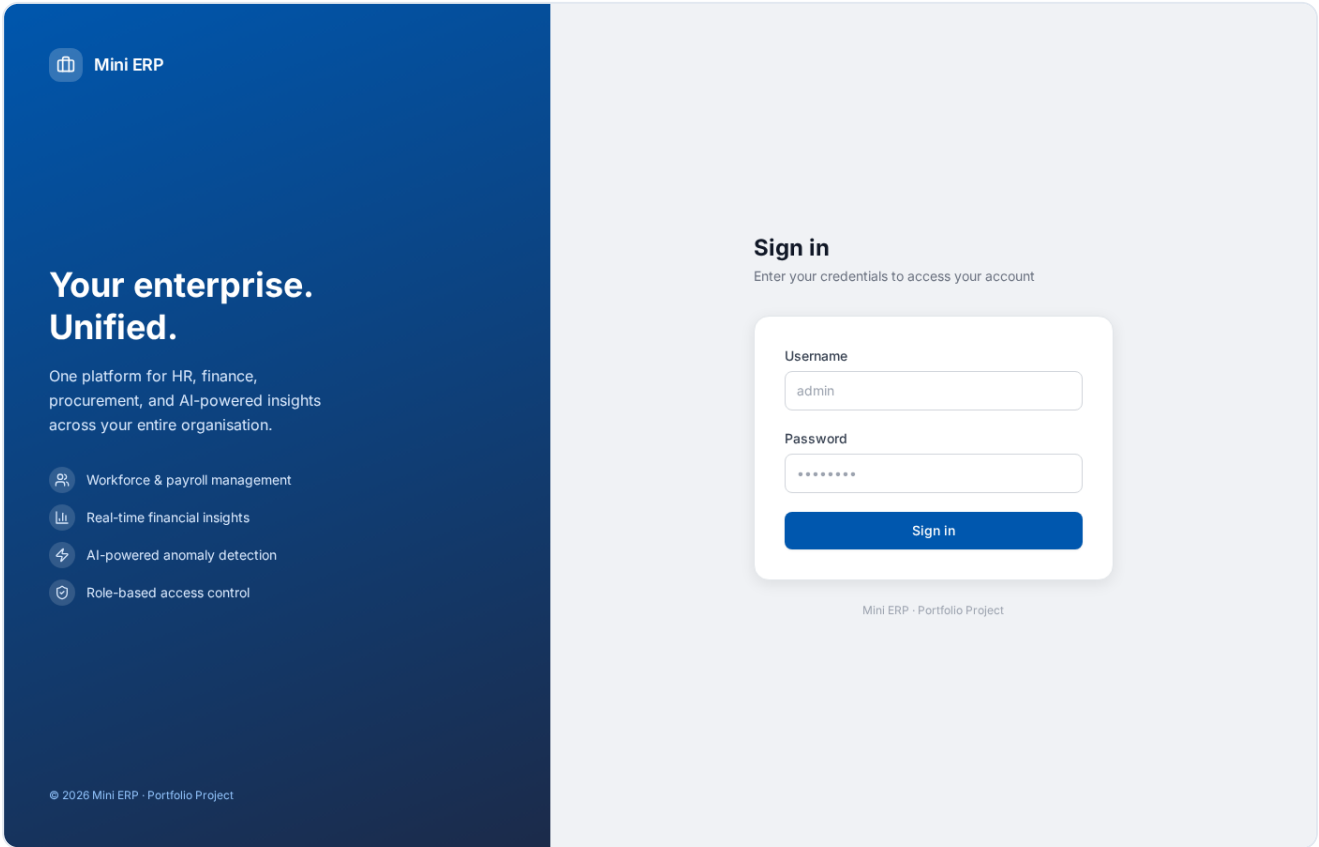
Database Schemas

Schema	Purpose
auth	Users, roles, permissions, refresh tokens
hcm	Departments, job families, jobs, employees
payroll	Pay groups, periods, runs, payslips, components
ap	Vendors, invoices, invoice lines, vouchers
expenses	Categories, expense reports, expense lines
procurement	Purchase requisitions, orders, PO lines, goods receipts
gl	Chart of accounts, fiscal periods, journals, budgets
mlops	ML models, metrics, prediction logs, training jobs, LLM conversations

3. Getting Started — Login

3.1 Accessing the Platform

Navigate to `http://<VM-IP>` in your browser. You will see the login page.



3.2 Authentication

Enter your credentials and click **Sign In**. The platform uses JWT-based authentication with automatic token refresh.

Default Roles

Role	Permissions
<code>superadmin</code>	Full access to all modules
<code>hr_manager</code>	Workforce + Payroll
<code>finance_manager</code>	AP + Expenses + GL + Procurement
<code>employee</code>	Own expense reports only

3.3 Session Management

- Access tokens expire after a configurable TTL (default: 30 minutes).
- Refresh tokens extend sessions automatically.
- Use the **Logout** option in the top-right user menu to terminate your session.

4. Week 0 — Infrastructure Setup

What was built

Component	Details
GCP VM	<code>e2-micro</code> (free tier), <code>us-central1</code> , 10 GB boot disk
GitHub	Repository with <code>main</code> / <code>develop</code> / <code>feature/*</code> branch strategy
CI/CD	GitHub Actions pipeline skeleton
Database	PostgreSQL 15 with 8 schema namespaces, Alembic migrations
Docker	Docker Compose stack: backend, frontend, db, redis, celery
Networking	Nginx reverse proxy, HTTP/HTTPS

Running the stack

```
# Start all services
docker compose -f docker-compose.dev.yml up -d --build

# Run DB migrations
docker compose -f docker-compose.dev.yml exec backend alembic upgrade head

# Seed test data
docker compose -f docker-compose.dev.yml exec backend python -m app.data.seed_all

# Health check
curl http://localhost:8000/api/v1/health
```

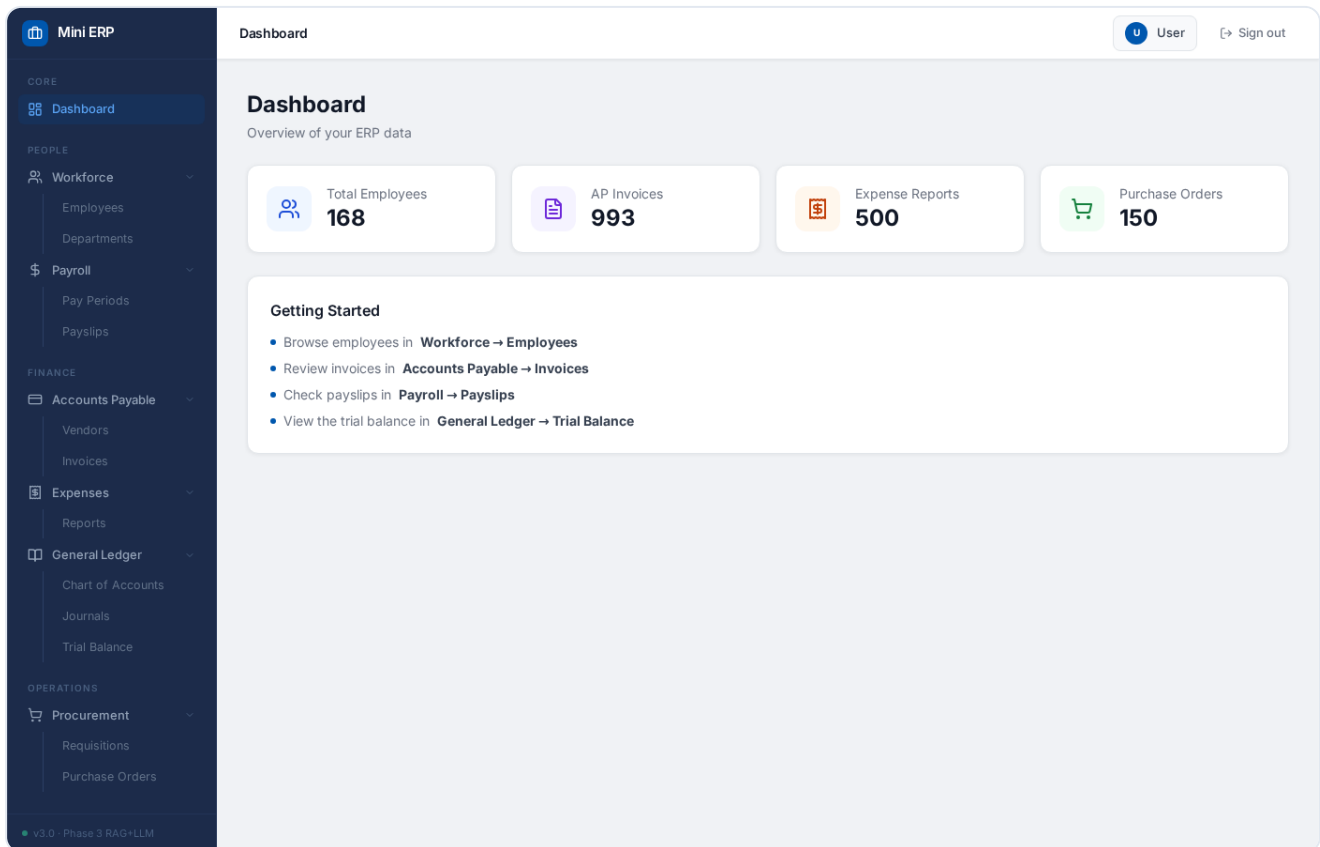
5. Phase 1 — Core ERP Modules

Release: v1.0.0 — 2026-02-XX

Phase 1 delivered a complete, production-quality ERP CRUD platform across six business modules, with JWT-based RBAC, full async FastAPI backend, React frontend, and a GCP Docker deployment.

5.1 Dashboard

The dashboard provides a real-time summary of key metrics across all ERP modules.



Metrics displayed:

- Active employees and recent hires
- Open AP invoices and outstanding balance
- Pending expense reports awaiting approval
- Open purchase requisitions and orders
- GL journal count and budget utilisation

5.2 Workforce (HCM)

Employees

Manage the full employee lifecycle: hire, update, and terminate.

Mini ERP

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v3.0 - Phase 3 RAG+LLM

Employees

168 total employees

Q Search by name or email...

ID	NAME	DEPARTMENT	JOB TITLE	HIRE DATE	STATUS	ATTRITION RISK
EMP-00009	Jordan Adams emp-00009@mini-erp.local	Marketing	Manager	Sep 27, 2023	active	10.0%
EMP-00040	Suzanne Adams emp-00040@mini-erp.local	Finance	Financial Accountant	Jun 1, 2020	active	9.6%
EMP-00192	Julian Ali emp-00192@mini-erp.local	Customer Success	Sales Executive	Jul 17, 2023	active	9.1%
EMP-00032	Alexandra Allen emp-00032@mini-erp.local	Software Development	Software Engineer	Apr 24, 2016	active	3.6%
EMP-00096	Graeme Allen emp-00096@mini-erp.local	General Ledger	Manager	Aug 12, 2019	active	5.3%
EMP-00045	Carole Armstrong emp-00045@mini-erp.local	Accounts Payable	AP Clerk	Feb 13, 2023	active	9.6%
EMP-00027	Mary Arnold emp-00027@mini-erp.local	Accounts Payable	Manager	Oct 4, 2018	active	6.5%
EMP-00101	Zoe Arnold emp-00101@mini-erp.local	Executive	Chief Executive Officer	Mar 23, 2023	active	12.2%
EMP-00047	Liam Bailey emp-00047@mini-erp.local	Warehouse	Manager	Nov 6, 2020	active	6.4%
EMP-00103	Norman Banks emp-00103@mini-erp.local	Accounts Payable	AP Clerk	Sep 12, 2020	active	7.0%
EMP-00195	Christine Barker	Legal	Director	Nov 4, 2021	active	8.3%

Key fields: Name, employee ID, department, job title, hire date, employment type, salary, status.

Actions:

- **Hire** — Create a new employee record linked to a department and job.
- **Edit** — Update personal info, job assignment, salary.
- **Terminate** — Sets status to **TERMINATED** and records termination date and reason.

Departments

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Departments

u User

Sign out

Departments

15 departments

CODE	NAME	DESCRIPTION	STATUS
FIN-AP	Accounts Payable	Processes vendor invoices, manages payment runs, and maintains supplier account records.	Active
CS	Customer Success	Manages post-sale customer relationships, onboarding, renewals, and satisfaction metrics.	Active
EXEC	Executive	Senior leadership responsible for company strategy, governance, and organisational direction.	Active
FIN	Finance	Oversees financial planning, reporting, treasury, and regulatory compliance.	Active
FIN-GL	General Ledger	Maintains the chart of accounts, journals, period close, and statutory reporting.	Active
HR	Human Resources	Responsible for talent acquisition, employee relations, performance management, and HR policy.	Active
IT	Information Technology	Owns the technology strategy, systems architecture, and digital transformation initiatives.	Active
IT-INF	Infrastructure	Manages cloud infrastructure, networking, security, and platform reliability.	Active
LEGAL	Legal	Handles contracts, regulatory compliance, corporate governance, and dispute resolution.	Active
MKT	Marketing	Drives brand strategy, demand generation, content, and go-to-market campaigns.	Active
OPS	Operations	Coordinates day-to-day business operations, process improvement, and cross-functional delivery.	Active
PROC	Procurement	Handles purchase requisitions, supplier sourcing, contract negotiation, and goods receipt.	Active
SALES	Sales	Responsible for new business development, account management, and revenue targets.	Active
IT-DEV	Software Development	Builds and maintains internal and customer-facing software products and services.	Active
OPS-WH	Warehouse	Manages goods receipt, stock control, picking, packing, and dispatch operations.	Active

Departments sit at the top of the HCM hierarchy. Each department contains job families → jobs → employees.

Hierarchy: Department → Job Family → Job → Employee

API Endpoints

Method	Endpoint	Description
GET	/api/v1/workforce/employees	List employees (paginated, filterable)
POST	/api/v1/workforce/employees	Create employee
GET	/api/v1/workforce/employees/{id}	Get employee detail
PUT	/api/v1/workforce/employees/{id}	Update employee
POST	/api/v1/workforce/employees/{id}/terminate	Terminate employee
GET	/api/v1/workforce/departments	List departments
POST	/api/v1/workforce/departments	Create department
GET	/api/v1/workforce/job-families	List job families
GET	/api/v1/workforce/jobs	List jobs

5.3 Payroll

Pay Periods

Mini ERP

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Pay Periods

36 pay periods

PERIOD	PAY GROUP	START	END	PAY DATE	STATUS
FY2026-P36	UK Monthly Payroll	Mar 1, 2026	Mar 31, 2026	Apr 28, 2026	closed
FY2026-P35	UK Monthly Payroll	Feb 1, 2026	Feb 28, 2026	Mar 28, 2026	closed
FY2026-P34	UK Monthly Payroll	Jan 1, 2026	Jan 31, 2026	Feb 28, 2026	closed
FY2025-P33	UK Monthly Payroll	Dec 1, 2025	Dec 31, 2025	Jan 28, 2026	closed
FY2025-P32	UK Monthly Payroll	Nov 1, 2025	Nov 30, 2025	Dec 28, 2025	closed
FY2025-P31	UK Monthly Payroll	Oct 1, 2025	Oct 31, 2025	Nov 28, 2025	closed
FY2025-P30	UK Monthly Payroll	Sep 1, 2025	Sep 30, 2025	Oct 28, 2025	closed
FY2025-P29	UK Monthly Payroll	Aug 1, 2025	Aug 31, 2025	Sep 28, 2025	closed
FY2025-P28	UK Monthly Payroll	Jul 1, 2025	Jul 31, 2025	Aug 28, 2025	closed
FY2025-P27	UK Monthly Payroll	Jun 1, 2025	Jun 30, 2025	Jul 28, 2025	closed
FY2025-P26	UK Monthly Payroll	May 1, 2025	May 31, 2025	Jun 28, 2025	closed
FY2025-P25	UK Monthly Payroll	Apr 1, 2025	Apr 30, 2025	May 28, 2025	closed
FY2025-P24	UK Monthly Payroll	Mar 1, 2025	Mar 31, 2025	Apr 28, 2025	closed
FY2025-P23	UK Monthly Payroll	Feb 1, 2025	Feb 28, 2025	Mar 28, 2025	closed
FY2025-P22	UK Monthly Payroll	Jan 1, 2025	Jan 31, 2025	Feb 28, 2025	closed

Pay periods define the payroll calendar. Each period belongs to a **pay group** (e.g., Monthly Salaried, Bi-Weekly Hourly) and has a status lifecycle:

OPEN → PROCESSING → COMPLETED → POSTED

Payslips

Mini ERP	Payslips U User Sign out				
CORE					
Dashboard					
PEOPLE					
Workforce					
Employees					
Departments					
\$ Payroll					
Pay Periods					
Payslips					
FINANCE					
Accounts Payable					
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Payslips
5692 payslips

EMPLOYEE	PERIOD	GROSS PAY	DEDUCTIONS	NET PAY	STATUS
Francesca Harrison	FY2023-P01	\$10,116.67	\$3,407.90	\$6,708.77	confirmed
Samuel Elliott	FY2023-P01	\$6,000.00	\$1,884.74	\$4,115.26	confirmed
Teresa Hurst	FY2023-P01	\$6,350.00	\$2,014.24	\$4,335.76	confirmed
Jessica White	FY2023-P01	\$4,300.00	\$1,255.74	\$3,044.26	confirmed
Duncan Humphries	FY2023-P01	\$7,383.33	\$2,396.58	\$4,986.75	confirmed
Kirsty Hawkins	FY2023-P01	\$6,375.00	\$2,023.49	\$4,351.51	confirmed
Carly Hale	FY2023-P01	\$4,666.67	\$1,391.40	\$3,275.27	confirmed
Ashley Bond	FY2023-P01	\$9,466.67	\$3,167.40	\$6,299.27	confirmed
Gail Oliver	FY2023-P01	\$11,600.00	\$3,956.74	\$7,643.26	confirmed
Jemma Cook	FY2023-P01	\$12,425.00	\$4,261.99	\$8,163.01	confirmed
Jack Griffiths	FY2023-P01	\$6,675.00	\$2,134.49	\$4,540.51	confirmed
Scott Knowles	FY2023-P01	\$3,425.00	\$931.99	\$2,493.01	confirmed
Mary Arnold	FY2023-P01	\$5,391.67	\$1,659.65	\$3,732.02	confirmed
Francesca Patel	FY2023-P01	\$5,233.33	\$1,601.08	\$3,632.25	confirmed
Declan Brown	FY2023-P01	\$12,808.33	\$4,403.83	\$8,404.50	confirmed

Payslips are generated per employee per pay run. Each payslip lists:

- **Earnings** — base salary, overtime, bonuses
- **Deductions** — taxes, benefits, retirement
- **Net pay**

Running payroll:

1. Create a pay period for the relevant pay group.
2. POST to `/api/v1/payroll/runs` to trigger payslip generation.
3. Payslips are generated asynchronously via Celery.
4. Review payslips, then post to the General Ledger.

API Endpoints

Method	Endpoint	Description
GET	/api/v1/payroll/pay-periods	List pay periods
POST	/api/v1/payroll/pay-periods	Create pay period
POST	/api/v1/payroll/runs	Trigger payroll run
GET	/api/v1/payroll/payslips	List payslips
GET	/api/v1/payroll/payslips/{id}	Get payslip detail
GET	/api/v1/payroll/components	List payroll components

5.4 Accounts Payable (AP)

Vendors

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Vendors

50 vendors

CODE	NAME	EMAIL	PHONE	TERMS	STATUS
VND - 00032	AI Innovations	accounts@aiinnovations.com	+44(0)292018878	Net 14	Active
VND - 00046	AuditExpert UK	accounts@auditexpertuk.com	02074960826	Net 30	Active
VND - 00050	BenefitsMax Ltd	accounts@benefitsmaxltd.com	(0161) 496 0146	Net 60	Active
VND - 00033	BlockChain Services	accounts@blockchainservices.com	+44151 496 0072	Net 60	Active
VND - 00040	BrandForge	accounts@brandforge.com	(0306) 999 0777	Net 30	Active
VND - 00004	BritConsult Partners	accounts@britconsultpartners.com	0808 1570975	Net 14	Active
VND - 00026	CloudInfra UK	accounts@cloudinfrauk.com	+441214960626	Net 30	Active
VND - 00003	CloudServ Solutions	accounts@cloudservsolutions.com	01214960510	Net 30	Active
VND - 00044	ComplianceFirst	accounts@compliancefirst.com	(0306) 9990996	Net 45	Active
VND - 00017	ConsultX Partners	accounts@consultxpartners.com	(0141) 496 0814	Net 45	Active
VND - 00041	ContentKing	accounts@contentking.com	029 2018 0700	Net 45	Active

Vendors are the supplier master. Each vendor has: name, tax ID, payment terms, contact info, and bank details.

Invoices

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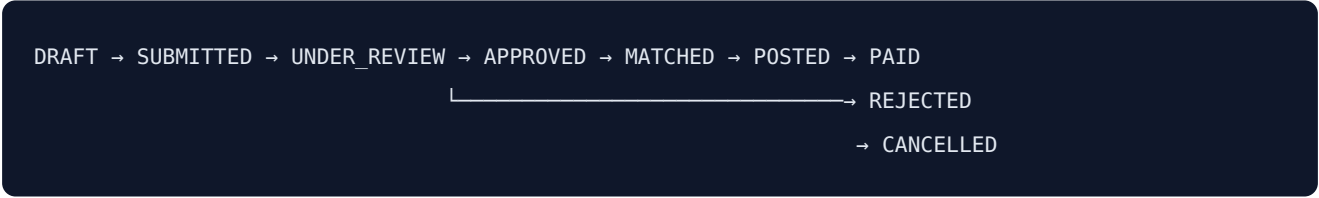
Invoices

993 invoices

All statuses

INVOICE #	VENDOR	DATE	DUE	AMOUNT	PAID	STATUS
INV-000002	SmartOffice	Sep 23, 2023	Oct 7, 2023	\$45,465.64	\$0.00	paid
INV-000003	TaxSavers Partners	Sep 19, 2025	Nov 18, 2025	\$22,747.16	\$0.00	paid
INV-000004	LegalEagle LLP	Oct 22, 2025	Nov 21, 2025	\$13,456.06	\$0.00	paid
INV-000005	DataEdge Analytics	May 18, 2024	Jun 17, 2024	\$51,180.89	\$0.00	paid
INV-000006	RapidRepair Services	Jul 21, 2024	Aug 4, 2024	\$19,985.57	\$0.00	paid
INV-000007	ContentKing	Aug 2, 2025	Sep 16, 2025	\$1,389.92	\$0.00	paid
INV-000008	DataVault Systems	Apr 9, 2026	May 24, 2026	\$54,740.56	\$0.00	submitted
INV-000009	SEO Masters UK	Jan 4, 2025	Feb 18, 2025	\$55,310.86	\$0.00	paid
INV-000010	SwiftSupply Co	Sep 15, 2025	Sep 29, 2025	\$57,393.22	\$0.00	paid
INV-000011	FinanceFirst Ltd	Jan 13, 2026	Feb 12, 2026	\$36,317.84	\$0.00	posted
INV-000012	ComplianceFirst	Jun 8, 2025	Jul 23, 2025	\$53,599.88	\$0.00	paid
INV-000013	UtilityFirst	Mar 17, 2024	Mar 31, 2024	\$46,650.80	\$0.00	paid
INV-000014	MediaMax Group	Jun 25, 2024	Aug 9, 2024	\$8,969.08	\$0.00	approved
INV-000015	SmartOffice	Jun 10, 2025	Jul 2, 2025	\$4,612.31	\$0.00	paid

Invoice status lifecycle:



Business rules for queries:

- **Outstanding** = status NOT IN (PAID , REJECTED , CANCELLED)
- **Unpaid/Open** = status IN (SUBMITTED , UNDER_REVIEW , APPROVED , MATCHED , POSTED)

Invoice lines support line-item detail. Vouchers record the payment event against an approved invoice.

API Endpoints

Method	Endpoint	Description
GET	/api/v1/ap/vendors	List vendors
POST	/api/v1/ap/vendors	Create vendor
GET	/api/v1/ap/invoices	List invoices (filterable by status, vendor)
POST	/api/v1/ap/invoices	Create invoice
PUT	/api/v1/ap/invoices/{id}/status	Update invoice status
GET	/api/v1/ap/vouchers	List payment vouchers
POST	/api/v1/ap/vouchers	Create voucher

5.5 Expenses

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Expense Reports

500 reports

All statuses

REPORT #	TITLE	EMPLOYEE	AMOUNT	SUBMITTED	STATUS
EXP-000002	Business expenses Jul 2025	Joanna Hall	\$205.21	Apr 12, 2026	under_review
EXP-000003	Business expenses Oct 2025	Damian Yates	\$292.89	Apr 12, 2026	approved
EXP-000004	Business expenses May 2025	Andrea Clarke	\$408.74	Apr 12, 2026	approved
EXP-000005	Business expenses Apr 2025	Chelsea Potts	\$801.96	Apr 12, 2026	submitted
EXP-000006	Business expenses Mar 2026	Ellie Perry	\$571.61	Apr 12, 2026	approved
EXP-000007	Business expenses Feb 2026	Anne Kirk	\$890.69	Apr 12, 2026	approved
EXP-000008	Business expenses Feb 2026	Janet Franklin	\$603.71	Apr 12, 2026	paid
EXP-000009	Business expenses Jul 2025	Denise Jones	\$355.18	Apr 12, 2026	paid
EXP-000010	Business expenses Apr 2025	Shirley Smith	\$60.14	Apr 12, 2026	submitted
EXP-000011	Business expenses Jun 2025	Callum Todd	\$478.91	Apr 12, 2026	under_review
EXP-000012	Business expenses Aug 2025	Charlene Rogers	\$131.17	Apr 12, 2026	approved
EXP-000013	Business expenses Oct 2025	Damian Yates	\$678.22	Apr 12, 2026	approved
EXP-000014	Business expenses Sep 2025	Jordan Adams	\$107.21	Apr 12, 2026	approved
EXP-000015	Business expenses May 2025	Jordan Duncan	\$102.21	Apr 12, 2026	under_review

Employees submit expense reports containing individual expense lines. Managers approve or reject at the report level.

Status lifecycle: DRAFT → SUBMITTED → APPROVED / REJECTED

Expense line fields: Date, category, amount, currency, description, receipt reference.

API Endpoints

Method	Endpoint	Description
GET	/api/v1/expenses/reports	List expense reports
POST	/api/v1/expenses/reports	Create expense report
POST	/api/v1/expenses/reports/{id}/submit	Submit for approval
POST	/api/v1/expenses/reports/{id}/approve	Approve report
POST	/api/v1/expenses/reports/{id}/reject	Reject report
GET	/api/v1/expenses/categories	List expense categories

5.6 Procurement

Purchase Requisitions

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Requisitions

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Purchase Requisitions

200 requisitions

All statuses

REQ #	TITLE	REQUESTER	AMOUNT	REQUIRED BY	STATUS
PR-00002	Innovate One-To-One Models	Martin Edwards	\$1,030.00	Mar 21, 2024	approved
PR-00003	Optimize Open-Source Initiatives	Terry Pugh	\$3,804.00	Oct 16, 2024	approved
PR-00004	Expedite Viral Methodologies	Sean Jennings	\$371.00	Apr 20, 2024	approved
PR-00005	Redefine Rich Web Services	Graeme Allen	\$179.00	Oct 17, 2024	submitted
PR-00006	Orchestrate B2B Vortals	Joanna Hall	\$1,034.00	Oct 29, 2024	approved
PR-00007	Optimize One-To-One Networks	Leigh Ross	\$203.00	Dec 21, 2024	submitted
PR-00008	Harness Rich E-Commerce	Jay Burns	\$1,858.00	Jan 10, 2025	approved
PR-00009	Deploy Rich Users	Hannah Brown	\$2,969.00	Dec 27, 2024	submitted
PR-00010	Exploit Magnetic Infrastructures	Geoffrey Davis	\$451.00	Mar 28, 2024	submitted
PR-00011	Enable One-To-One Markets	Christine Barker	\$3,928.00	Sep 5, 2024	submitted
PR-00012	Engage Next-Generation Action-Items	Ellie Perry	\$1,862.00	Oct 9, 2024	submitted
PR-00013	Synergize Proactive E-Commerce	Michael Johnson	\$783.00	Mar 11, 2024	approved
PR-00014	Orchestrate Bricks-And-Clicks Eyeballs	Jennifer Lee	\$3,936.00	Sep 8, 2024	submitted
PR-00015	Engineer Viral Info-Mediascapes	Gleno Martin	\$157.00	Nov 14, 2024	approved

Employees raise purchase requisitions (PRs) to request goods or services. PRs are reviewed and converted to Purchase Orders upon approval.

Purchase Orders

Mini ERP CORE Dashboard PEOPLE Workforce Employees Departments Payroll Pay Periods Payslips FINANCE Accounts Payable Vendors Invoices Expenses Reports General Ledger Chart of Accounts Journals Trial Balance OPERATIONS Procurement Requisitions Purchase Orders v3.0 - Phase 3 RAG+LLM	Purchase Orders					User Sign out	
	Purchase Orders					150 purchase orders	
	All statuses						
	P0 #	VENDOR	ORDER DATE	DELIVERY	AMOUNT	STATUS	
	P0-00002	FinanceFirst Ltd	May 28, 2024	Jun 8, 2024	\$399.90	issued	
	P0-00003	PrintCraft UK	Dec 29, 2024	Jan 18, 2025	\$3,984.00	closed	
	P0-00004	MarketBoost Agency	Jun 6, 2024	Jul 2, 2024	\$3,735.00	closed	
	P0-00005	MarketBoost Agency	Jun 15, 2024	Jul 18, 2024	\$1,089.00	partial	
	P0-00006	SwiftSupply Co	Dec 19, 2024	Jan 29, 2025	\$7,192.00	partial	
	P0-00007	ComplianceFirst	Dec 10, 2024	Jan 18, 2025	\$979.00	closed	
	P0-00008	LegalEagle LLP	Oct 14, 2024	Oct 27, 2024	\$1,396.00	partial	
	P0-00009	BlockChain Services	Aug 26, 2024	Sep 27, 2024	\$6,293.00	closed	
	P0-00010	DevOps Masters	Jul 19, 2024	Aug 25, 2024	\$890.00	issued	
	P0-00011	BlockChain Services	May 6, 2024	May 30, 2024	\$1,513.00	issued	
	P0-00012	TechVision Ltd	Dec 16, 2024	Jan 9, 2025	\$4,980.00	issued	
	P0-00013	SoftwarePlus Ltd	Apr 21, 2024	May 10, 2024	\$396.00	partial	
	P0-00014	AuditExpert UK	Jun 24, 2024	Jul 27, 2024	\$27,489.00	issued	
	P0-00015	DataEdge Analytics	Jun 24, 2024	Jul 24, 2024	\$100.00	partial	

Purchase Orders (POs) are sent to vendors. Goods Receipts record delivery against a PO line.

Status lifecycle (PO): DRAFT → APPROVED → SENT → PARTIALLY_RECEIVED → RECEIVED → CLOSED

API Endpoints

Method	Endpoint	Description
GET	/api/v1/procurement/requisitions	List purchase requisitions
POST	/api/v1/procurement/requisitions	Create requisition
POST	/api/v1/procurement/requisitions/{id}/approve	Approve requisition
GET	/api/v1/procurement/orders	List purchase orders
POST	/api/v1/procurement/orders	Create purchase order
POST	/api/v1/procurement/receipts	Record goods receipt

5.7 General Ledger (GL)

Chart of Accounts

Mini ERP

CORE

- Dashboard

PEOPLE

- Workforce
 - Employees
 - Departments
- Payroll
 - Pay Periods
 - Payslips

FINANCE

- Accounts Payable
 - Vendors
 - Invoices
- Expenses
 - Reports
- General Ledger
 - Chart of Accounts
 - Journals
 - Trial Balance

OPERATIONS

- Procurement
 - Requisitions
 - Purchase Orders

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Chart of Accounts

u User Sign out

Chart of Accounts

28 accounts

NUMBER	NAME	TYPE	NORMAL BALANCE	STATUS
1000	Current Assets	asset	debit	Active
1100	Cash and Bank	asset	debit	Active
1200	Accounts Receivable	asset	debit	Active
1300	Prepaid Expenses	asset	debit	Active
2000	Current Liabilities	liability	credit	Active
2100	Income Tax Payable	liability	credit	Active
2110	NI Payable	liability	credit	Active
2120	Pension Payable	liability	credit	Active
2200	Accounts Payable	liability	credit	Active
3000	Equity	equity	credit	Active
3100	Retained Earnings	equity	credit	Active
4000	Revenue	revenue	credit	Active
4100	Product Revenue	revenue	credit	Active
4200	Service Revenue	revenue	credit	Active
5000	Cost of Goods Sold	expense	debit	Active

The chart of accounts defines the financial structure. Account types: ASSET , LIABILITY , EQUITY , REVENUE , EXPENSE .

Journals

Mini ERP

CORE

- Dashboard

PEOPLE

- Workforce
 - Employees
 - Departments
- Payroll
 - Pay Periods
 - Payslips

FINANCE

- Accounts Payable
 - Vendors
 - Invoices
- Expenses
 - Reports
- General Ledger
 - Chart of Accounts
 - Journals
 - Trial Balance

OPERATIONS

- Procurement
 - Requisitions
 - Purchase Orders

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Journals

Journal Entries

54 Journals

JOURNAL #	DATE	DESCRIPTION	REFERENCE	DEBITS	CREDITS	STATUS
JNL-000054	Mar 31, 2026	Operating expenses — Mar 2026	accounts_payable	\$15,635.54	\$15,635.54	posted
JNL-000053	Mar 31, 2026	Payroll journal — Mar 2026	payroll	\$90,063.68	\$90,063.68	posted
JNL-000052	Feb 28, 2026	Operating expenses — Feb 2026	accounts_payable	\$5,429.04	\$5,429.04	posted
JNL-000051	Feb 28, 2026	Payroll journal — Feb 2026	payroll	\$90,513.80	\$90,513.80	posted
JNL-000050	Jan 31, 2026	Operating expenses — Jan 2026	accounts_payable	\$7,240.77	\$7,240.77	posted
JNL-000049	Jan 31, 2026	Payroll journal — Jan 2026	payroll	\$84,339.77	\$84,339.77	posted
JNL-000048	Dec 31, 2025	Operating expenses — Dec 2025	accounts_payable	\$5,396.36	\$5,396.36	posted
JNL-000047	Dec 31, 2025	Payroll journal — Dec 2025	payroll	\$89,672.58	\$89,672.58	posted
JNL-000045	Nov 30, 2025	Payroll journal — Nov 2025	payroll	\$119,992.98	\$119,992.98	posted
JNL-000046	Nov 30, 2025	Operating expenses — Nov 2025	accounts_payable	\$21,415.22	\$21,415.22	posted
JNL-000043	Oct 31, 2025	Payroll journal — Oct 2025	payroll	\$82,224.39	\$82,224.39	posted
JNL-000044	Oct 31, 2025	Operating expenses — Oct 2025	accounts_payable	\$6,779.24	\$6,779.24	posted
JNL-000042	Sep 30, 2025	Operating expenses — Sep 2025	accounts_payable	\$6,754.40	\$6,754.40	posted
JNL-000041	Sep 30, 2025	Payroll journal — Sep 2025	payroll	\$112,004.02	\$112,004.02	posted
JNL-000040	Aug 31, 2025	Operating expenses — Aug 2025	accounts_payable	\$5,748.46	\$5,748.46	posted

Journal entries record financial transactions as double-entry bookkeeping (debits = credits). Each journal has a status: DRAFT → POSTED.

Trial Balance

Mini ERP

CORE

Dashboard

PEOPLE

Workforce

Employees

Departments

\$ Payroll

Pay Periods

Payslips

FINANCE

Accounts Payable

Vendors

Invoices

Expenses

Reports

General Ledger

Chart of Accounts

Journals

Trial Balance

OPERATIONS

Procurement

Requisitions

Purchase Orders

v3.0 - Phase 3 RAG+LLM

Trial Balance

As of current date

Out of Balance

ACCOUNT	NAME	TYPE	DEBIT	CREDIT
1100	Cash and Bank	ASSET	—	\$180,143.72
2100	Income Tax Payable	LIABILITY	—	\$52,983.45
2110	NI Payable	LIABILITY	—	\$31,790.07
2200	Accounts Payable	LIABILITY	—	\$28,305.35
6100	Salaries — Basic Pay	EXPENSE	\$264,917.25	—
6500	Travel & Entertainment	EXPENSE	\$21,064.58	—
6800	Utilities	EXPENSE	\$7,240.77	—
Totals			\$293,222.60	\$293,222.59

The trial balance aggregates all posted journal lines per account and verifies that total debits equal total credits.

GL API Endpoints

Method	Endpoint	Description
GET	/api/v1/gl/accounts	List chart of accounts
POST	/api/v1/gl/accounts	Create GL account
GET	/api/v1/gl/fiscal-periods	List fiscal periods
GET	/api/v1/gl/journals	List journals
POST	/api/v1/gl/journals	Create journal entry
POST	/api/v1/gl/journals/{id}/post	Post journal to GL
GET	/api/v1/gl/trial-balance	Get trial balance report
GET	/api/v1/gl/budgets	List budgets
POST	/api/v1/gl/budgets	Create budget

5.8 Admin

Mini ERP

Users

9 users

USERNAME	FULL NAME	EMAIL	ROLES	STATUS	CREATED
workforce_user1	Alice Workforce	workforce1@mini-erp.local	workforce_user	Active	—
payroll_user1	Bob Payroll	payroll1@mini-erp.local	payroll_user	Active	—
ap_user1	Carol AP	ap1@mini-erp.local	ap_user	Active	—
expenses_user1	Dave Expenses	expenses1@mini-erp.local	expenses_user	Active	—
procurement_user1	Eve Procurement	procurement1@mini-erp.local	procurement_user	Active	—
gl_user1	Frank GL	gl1@mini-erp.local	gl_user	Active	—
hr_admin	Sarah HR Admin	hr.admin@mini-erp.local	hr_superuser	Active	—
finance_admin	James Finance Admin	fin.admin@mini-erp.local	finance_superuser	Active	—
platform_admin	Platform Admin	admin@mini-erp.local	platform_admin	Active	—

The Admin module provides user management and RBAC configuration.

Features:

- View all platform users and their assigned roles.
- Assign or revoke roles from users.
- View platform-wide statistics (user counts, module activity).

RBAC model: User → Roles → Permissions

Each permission is a `module:action` pair, e.g., `AP:CREATE`, `PAYROLL:APPROVE`, `AI:READ`.

API Endpoints

Method	Endpoint	Description
GET	/api/v1/auth/users	List all users
POST	/api/v1/admin/roles	Create role
POST	/api/v1/admin/users/{id}/roles	Assign role to user
DELETE	/api/v1/admin/users/{id}/roles/{role_id}	Revoke role from user
GET	/api/v1/admin/stats	Platform-wide statistics

6. Phase 2 — Machine Learning

Release: v2.0.0 — 2026-03-XX

Phase 2 added four ML models trained on ERP data, a model registry, prediction logging, and an AI Dashboard.

Models Overview

Model	Algorithm	Task
Attrition Predictor	RandomForest + SMOTE	Predict employee churn risk
Expense Violation Detector	GradientBoosting / IsolationForest	Flag policy-violating expense lines
Payroll Anomaly Detector	RandomForest / IsolationForest	Detect anomalous payslips
Invoice Classifier	RandomForest + TF-IDF	Categorise invoices into 10 spend categories

All models are:

- Saved as `.joblib` files to `MODELS_DIR=/app/models_store`
- Registered in the `mlops.ml_models` database table
- Versioned with evaluation metrics (accuracy, ROC-AUC, F1, CV-AUC)
- Logged per prediction to `mlops.ml_prediction_logs`

6.1 AI Insights

The screenshot displays the Mini ERP AI / MLOps interface. On the left is a dark sidebar with navigation links under categories: CORE (Dashboard), PEOPLE (Workforce, Employees, Departments), FINANCE (Payroll, Pay Periods, Payslips, Accounts Payable, Vendors, Invoices, Expenses, Reports, General Ledger, Chart of Accounts, Journals, Trial Balance), and OPERATIONS (Procurement, Requisitions, Purchase Orders). The main content area is titled 'AI / MLOps' and includes a subtitle 'Phase 2 ML — model registry, top-10 insights, and prediction audit log'. Below this are tabs for 'Model Registry', 'Insights' (selected), and 'Prediction Log'. The 'Insights' tab shows a 'Top 10 — Attrition Risk' report with a description: 'Predicts the likelihood of an employee leaving the organisation, based on tenure, performance rating, satisfaction score, salary-to-band ratio, and overtime patterns. Scores above 60% warrant a retention conversation.' The report includes a table with columns: #, EMPLOYEE, DEPARTMENT, and ATTRITION RISK. Each row shows an employee's name, ID, department, and a risk score with a corresponding green progress bar.

#	EMPLOYEE	DEPARTMENT	ATTRITION RISK
1	Martyn Barrett EMP-00178	Legal	28.2%
2	Annette Hughes EMP-00133	Warehouse	26.2%
3	Chloe Rhodes EMP-00165	Infrastructure	21.4%
4	Vanessa Blake EMP-00196	Operations	21.0%
5	Abigail Webb EMP-00115	Infrastructure	20.9%
6	Mohammad Wright EMP-00126	Legal	20.8%
7	Jade Robson EMP-00168	General Ledger	20.5%
8	Glenn Martin EMP-00052	Accounts Payable	19.2%
9	Jane Green EMP-00139	Customer Success	18.5%

The AI Insights tab provides batch predictions across all four ML models. Key views:

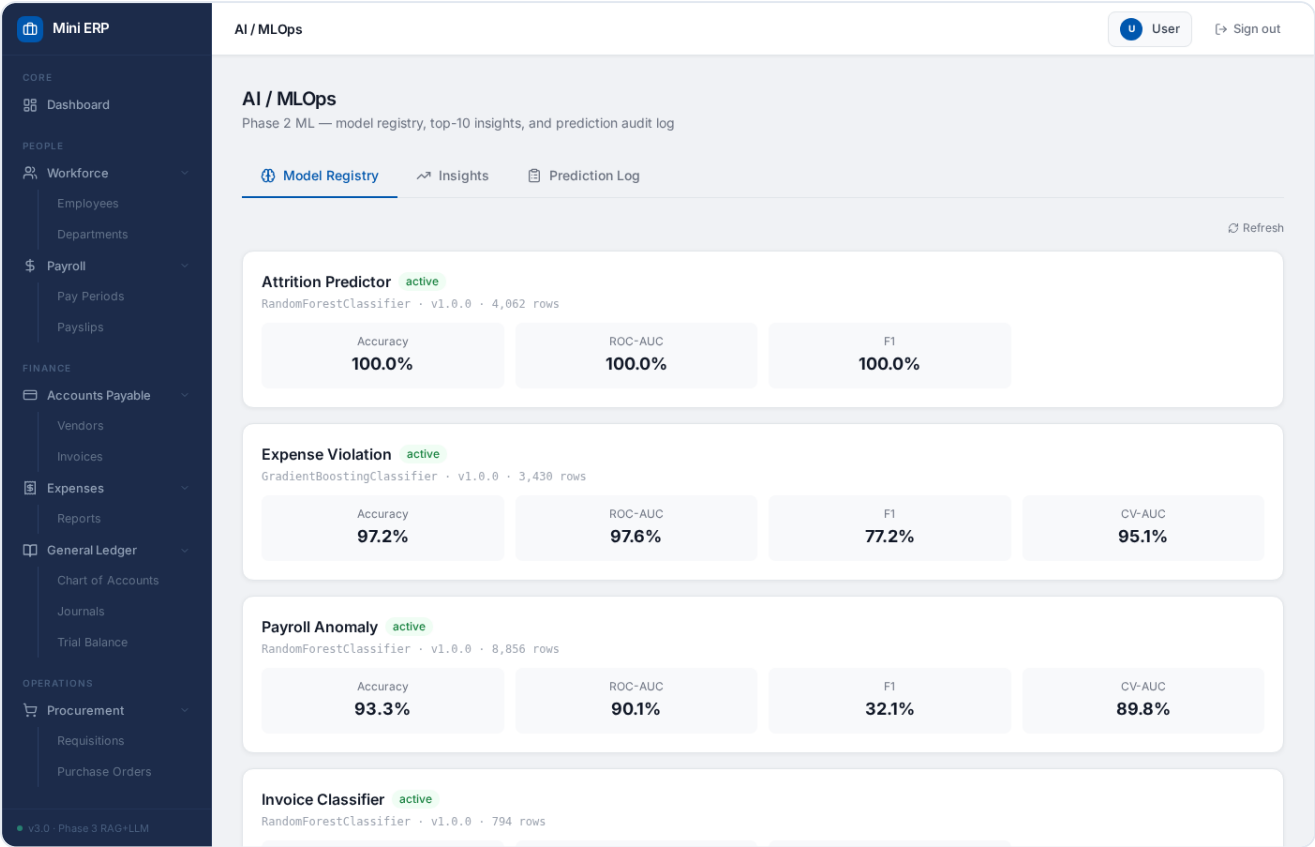
- **Attrition Risk** — Employee list with churn probability scores; highlights high-risk employees.
- **Expense Violations** — Expense lines flagged for potential policy violations with violation probability.
- **Payroll Anomalies** — Payslips with anomaly scores outside normal distribution.
- **Invoice Categories** — Invoices with AI-assigned spend categories and confidence scores.

Training a model (first-time or retraining):

```
POST /api/v1/mlops/train/attrition
POST /api/v1/mlops/train/expense_violation
POST /api/v1/mlops/train/payroll_anomaly
POST /api/v1/mlops/train/invoice_classifier
```

Training runs asynchronously via Celery. Monitor progress via the Model Registry.

6.2 Model Registry



The Model Registry lists all trained model versions with:

- Model name and version
- Training date and status (`TRAINING` , `ACTIVE` , `ARCHIVED`)
- Evaluation metrics: accuracy, ROC-AUC, F1 score, CV-AUC
- Feature importance (where applicable)

API Endpoints

Method	Endpoint	Description
GET	/api/v1/mlops/models	List all registered models
GET	/api/v1/mlops/models/{id}	Get model detail + metrics
POST	/api/v1/mlops/train/{model_type}	Trigger model training
GET	/api/v1/mlops/training-jobs	List training jobs

6.3 Prediction Log

The screenshot shows the Mini ERP interface with a dark sidebar on the left containing navigation links for CORE (Dashboard), PEOPLE (Workforce, Employees, Departments), FINANCE (Payroll, Pay Periods, Payslips, Accounts Payable, Vendors, Invoices, Expenses, Reports, General Ledger, Chart of Accounts, Journals, Trial Balance), and OPERATIONS (Procurement, Requisitions, Purchase Orders). The main content area is titled 'AI / MLOps' and includes a subtitle 'Phase 2 ML — model registry, top-10 insights, and prediction audit log'. Below this, there are tabs for 'Model Registry', 'Insights', and 'Prediction Log', with the latter being selected. A message states '1 predictions logged' with a 'Refresh' button. A table displays the prediction data:

MODEL ID	ENTITY	PREDICTION	PROBABILITY	LATENCY	TIME
92c71c0d...	employee/55df4084...	low_risk	15.6%	225ms	19 Apr 05:58

At the bottom left of the sidebar, it says 'v3.0 · Phase 3 RAG+LLM'. The top right of the main area shows a user profile 'User' and a 'Sign out' link.

Every prediction made by any ML model is logged to `mlops.ml_prediction_logs` with:

- Model name and version
- Input features (hashed/summarised)
- Predicted class and confidence score
- Timestamp and requesting user

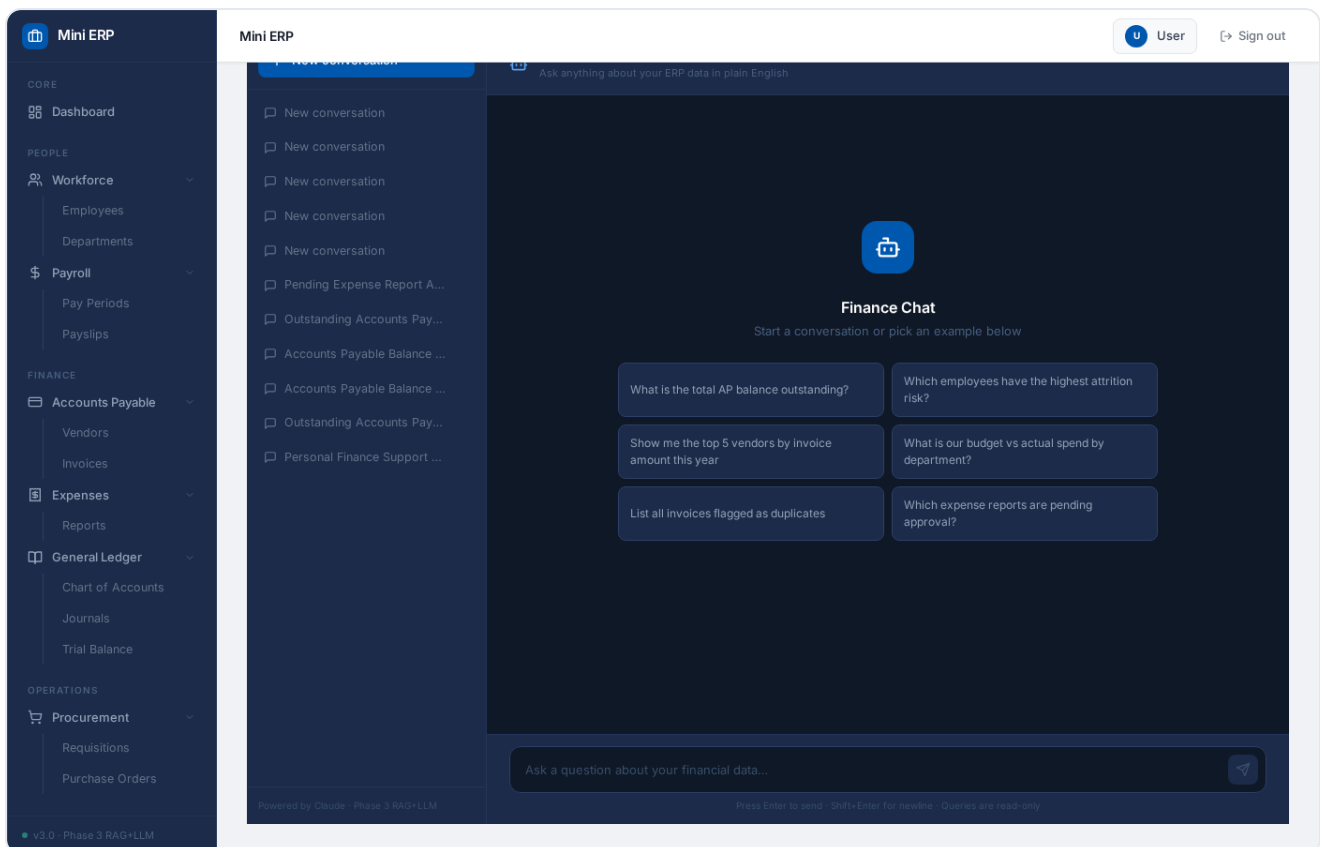
This provides a complete audit trail for all AI decisions made by the platform.

7. Phase 3 — RAG + LLM

Release: v3.1.0 — 2026-04-15 | **Bug fixes:** v3.2.0 — 2026-04-20

Phase 3 added natural-language querying via Text-to-SQL and vector-similarity-based duplicate invoice detection.

7.1 Finance Chat



Finance Chat lets users query the live ERP database using plain English. The system translates questions into SQL, executes them safely, and returns natural-language answers.

How It Works

```
Browser (ChatPage)
  | POST /api/v1/llmops/chat/conversations/{id}/message
  ▼
FastAPI (llmops.py router)
  | ai_service.send_message()
  ▼
chat_service.py
  |— Load conversation history
  |— LLM call 1 → may call run_sql tool
  |   |— SQL validated → SAVEPOINT → execute → markdown table
  |— LLM call 2 → natural-language answer (never raw JSON)
```

Using Finance Chat

1. Navigate to **AI Chat** in the left sidebar.
2. The welcome screen shows 6 example prompts — click any to start immediately.

3. Type your question in the input box and press **Enter**.
4. If no conversation is active, one is created automatically.
5. The assistant's response appears with an optional collapsible **SQL Query** panel showing:
 6. The generated SQL statement
 7. A preview of up to 20 result rows
 8. Previous conversations are listed in the left sidebar.

Example Queries

Question	What it does
What is the total AP outstanding balance?	Sums open invoices
Show department-wise AP outstanding balance	Joins invoices → departments
Which employees have the highest attrition risk?	Queries attrition risk scores
Top 5 vendors by invoice amount this year	Groups by vendor, filters by date
List all invoices flagged as duplicates	Filters <code>is_duplicate = true</code>
Which expense reports are pending approval?	Filters by status
Budget vs actual spend by department	Joins GL budgets → departments

LLM Provider Configuration

Set `LLM_PROVIDER` in `backend/.env.dev` :

Provider	<code>LLM_PROVIDER</code> value	Required config
Anthropic Claude	<code>anthropic</code>	<code>ANTHROPIC_API_KEY</code>
Groq (free tier)	<code>groq</code>	<code>GROQ_API_KEY</code> , <code>GROQ_CHAT_MODEL</code>
Ollama (local)	<code>ollama</code>	<code>OLLAMA_BASE_URL</code> , <code>OLLAMA_CHAT_MODEL</code>

SQL Safety

All generated queries are validated before execution:

- **Forbidden:** `INSERT` , `UPDATE` , `DELETE` , `DROP` , `CREATE` , `ALTER` , `TRUNCATE` , `GRANT` , `REVOKE`
- **Required:** must contain `SELECT`
- **Row limit:** every query is wrapped with `LIMIT <MAX_SQL_ROWS>` (default: 100)
- **Savepoint isolation:** each query runs inside a PostgreSQL `SAVEPOINT` — a failed query rolls back only the savepoint, not the whole session

API Endpoints

Method	Endpoint	Description
POST	/api/v1/llmops/chat/conversations	Create conversation
GET	/api/v1/llmops/chat/conversations	List conversations (50 most recent)
GET	/api/v1/llmops/chat/conversations/{id}	Get conversation + full message history
DELETE	/api/v1/llmops/chat/conversations/{id}	Archive conversation
POST	/api/v1/llmops/chat/conversations/{id}/message	Send message, receive LLM reply

Known Limitations

- Read-only access — no write operations via chat.
- Conversation history sent to LLM is capped at `CHAT_HISTORY_LIMIT` messages (default: 20).
- Groq/Ollama (Llama family) may emit tool calls as inline text — handled automatically by the backend.

7.2 Duplicate Invoice Detection

Invoice embeddings are generated using a text embedding model and stored in PostgreSQL via `pgvector`. Cosine similarity is used to detect near-duplicate invoices.

How it works:

1. When an invoice is created or updated, an embedding is generated from its key fields (vendor, amount, description, date).
2. The embedding is stored in `mlops.invoice_embeddings`.
3. A duplicate check compares the new embedding against all existing embeddings using cosine similarity.
4. Invoices above the similarity threshold are flagged as `is_duplicate = true`.

API Endpoints

Method	Endpoint	Description
POST	/api/v1/llmops/invoices/{id}/duplicate-check	Check single invoice for duplicates
GET	/api/v1/llmops/invoices/duplicates	List all flagged duplicate invoices
POST	/api/v1/llmops/embeddings/reindex	Reindex all invoice embeddings (admin)

8. API Reference

All endpoints require:

- Authorization: Bearer <access_token> header
- Appropriate RBAC permission for the module

Base URL: `http://<VM-IP>:8000/api/v1`

Interactive docs: `http://<VM-IP>:8000/docs` (Swagger UI)

Authentication

Method	Endpoint	Description
POST	/auth/login	Obtain access + refresh tokens
POST	/auth/refresh	Refresh access token
POST	/auth/logout	Invalidate refresh token
GET	/auth/me	Get current user profile
GET	/auth/users	List all users (admin)

Health Check

```
curl http://<VM-IP>:8000/api/v1/health
# → {"status": "healthy", "version": "3.2.0"}
```

9. Troubleshooting

Symptom	Cause	Fix
<code>psql: no socket</code>	Postgres is in Docker	Use <code>docker compose exec db psql ...</code>
<code>422 Unprocessable Entity</code>	Missing required field in request body	Check Pydantic schema in API docs
<code>403 Forbidden</code>	User lacks required RBAC permission	Assign correct role in Admin module
<code>relation does not exist</code>	Migrations not run	<code>alembic upgrade head</code>
ML model not found	Model not trained yet	<code>POST /api/v1/mlops/train/{model_type}</code>
Finance Chat: input frozen	(Fixed in v3.2.0)	Upgrade to v3.2.0
Finance Chat: raw JSON in response	(Fixed in v3.2.0)	Upgrade to v3.2.0
Finance Chat: <code>InFailedSQLTransactionError</code>	(Fixed in v3.2.0)	Upgrade to v3.2.0 — SQL now runs inside SAVEPOINT
Groq tool calls not working	Llama inline tool-call format	Handled automatically in v3.2.0+
Docker service won't start	Port conflict or volume issue	<code>docker compose down -v && docker compose up -d --build</code>

Useful Commands

```
# View backend logs
docker compose -f docker-compose.dev.yml logs -f backend

# View DB logs
docker compose -f docker-compose.dev.yml logs -f db

# Restart a single service
docker compose -f docker-compose.dev.yml restart backend

# Open a DB shell
docker compose -f docker-compose.dev.yml exec db psql -U erp_user -d mini_erp

# Re-run migrations
docker compose -f docker-compose.dev.yml exec backend alembic upgrade head

# Re-seed data
docker compose -f docker-compose.dev.yml exec backend python -m app.data.seed_all
```